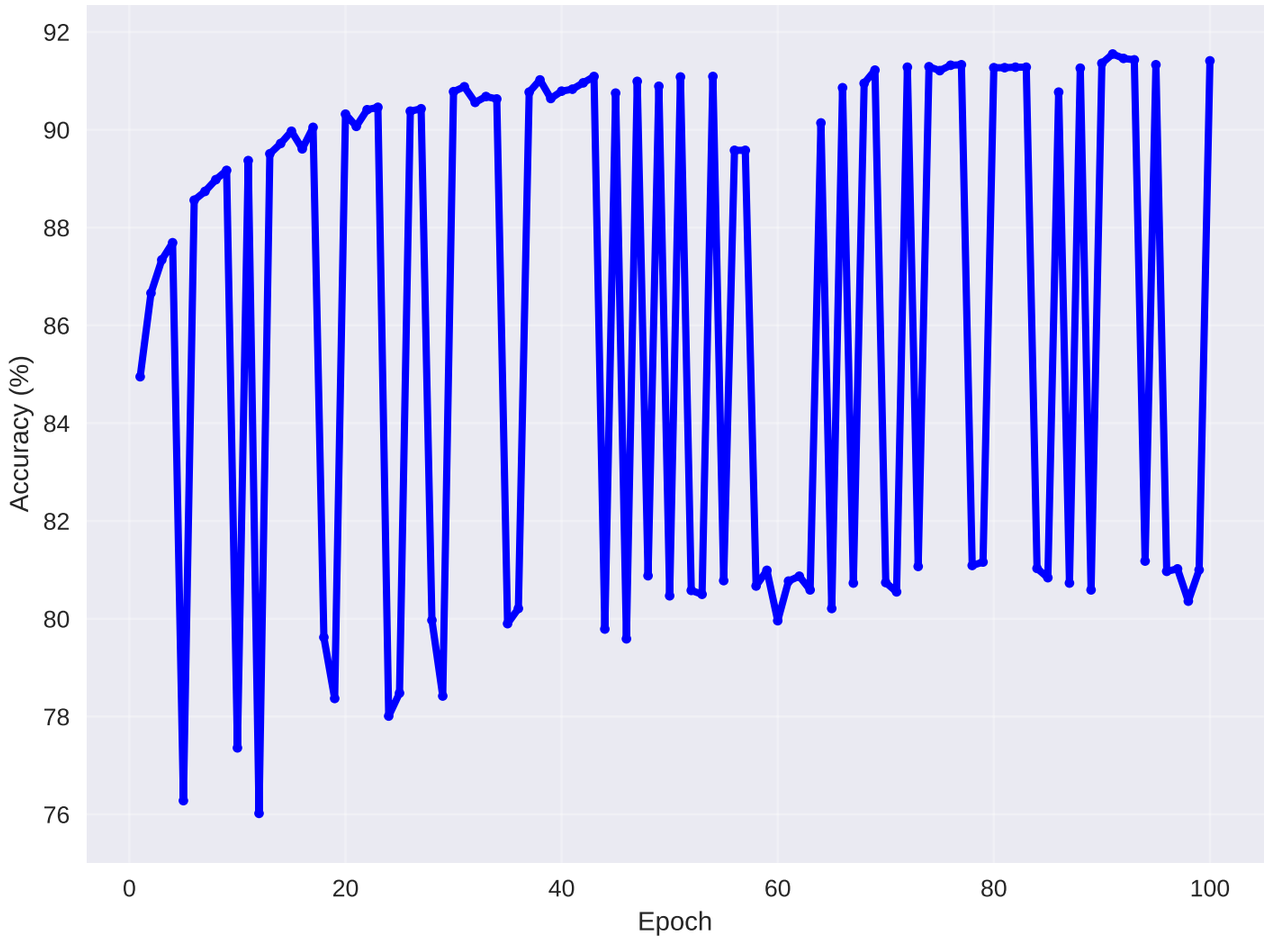
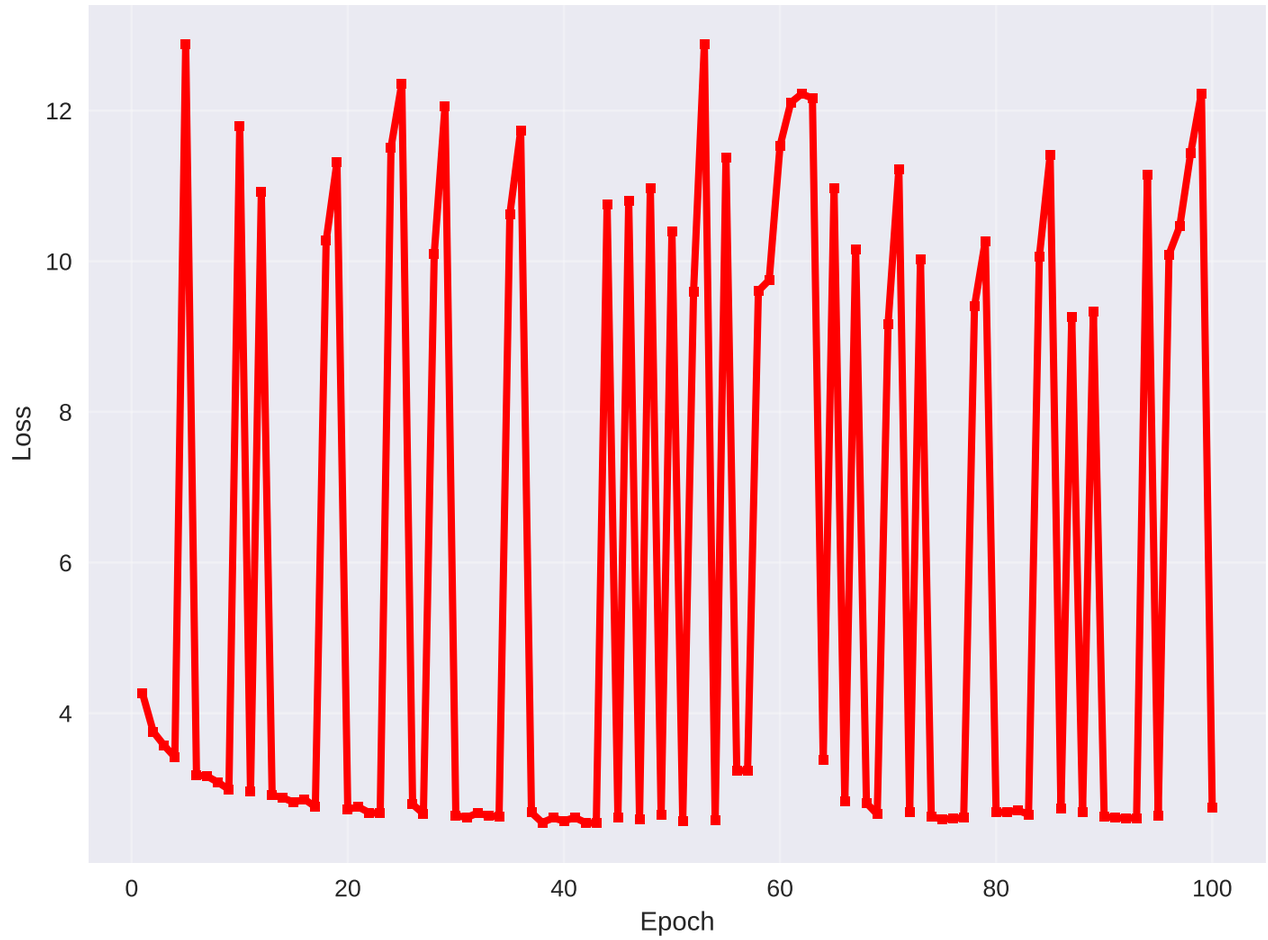
Federated Learning - Main Metrics Overview

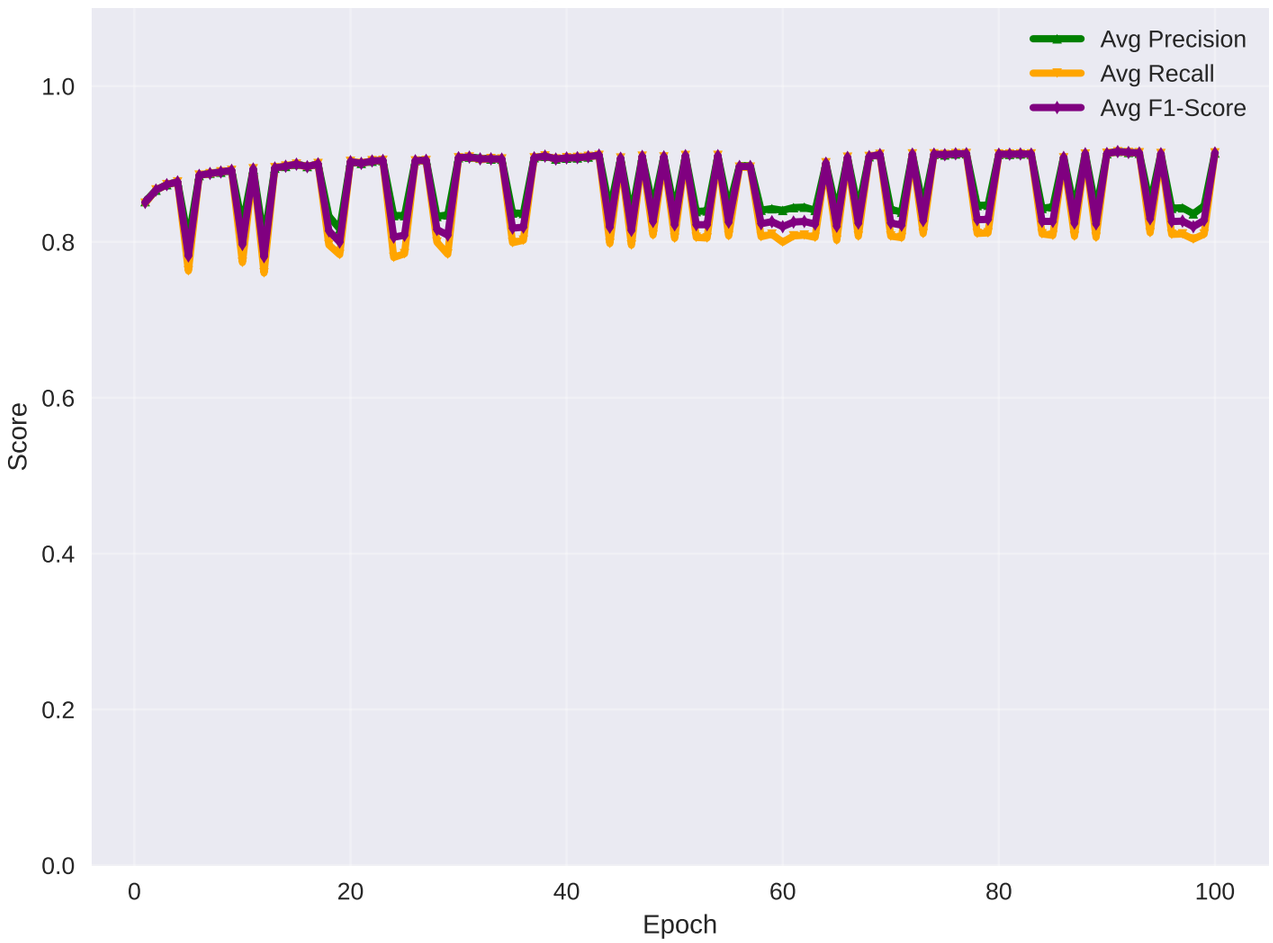
 Global Model Accuracy Evolution



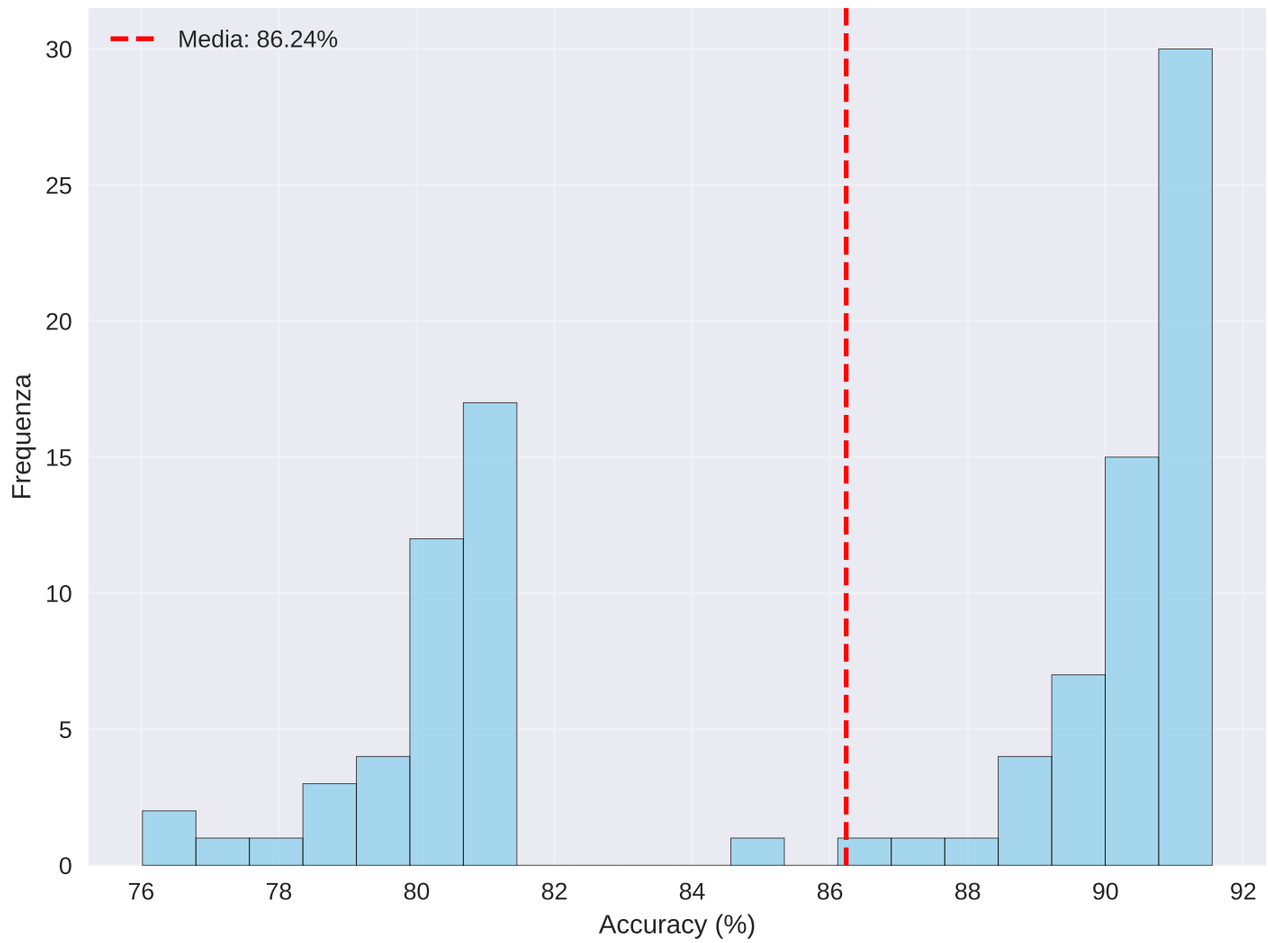
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

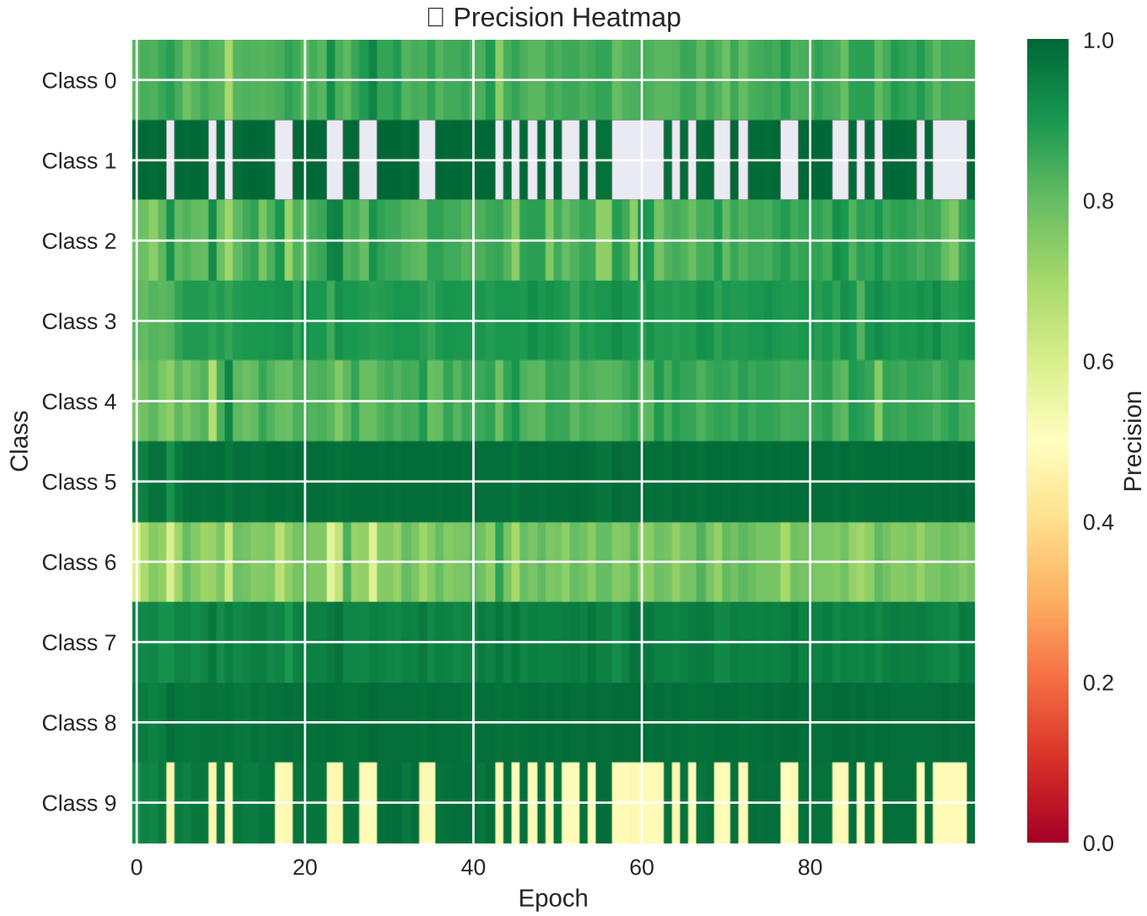


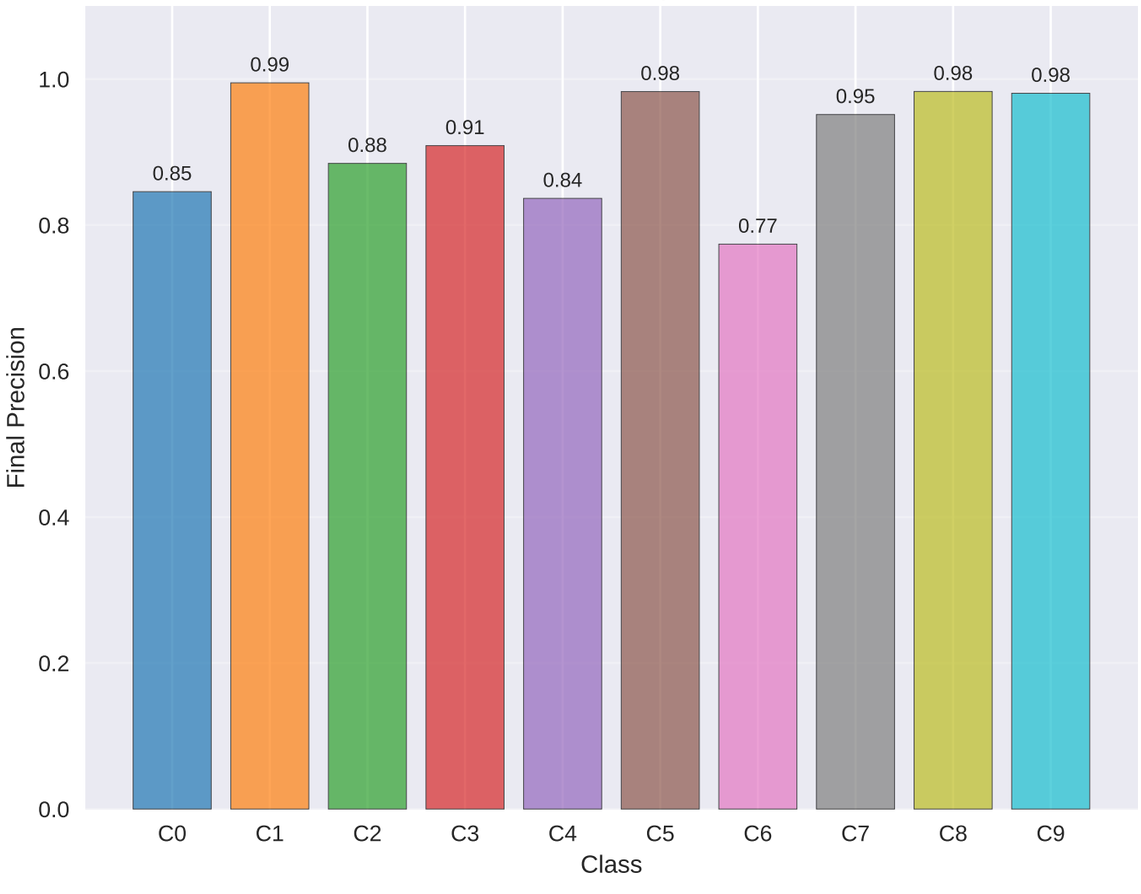
 Distribuzione Accuracy

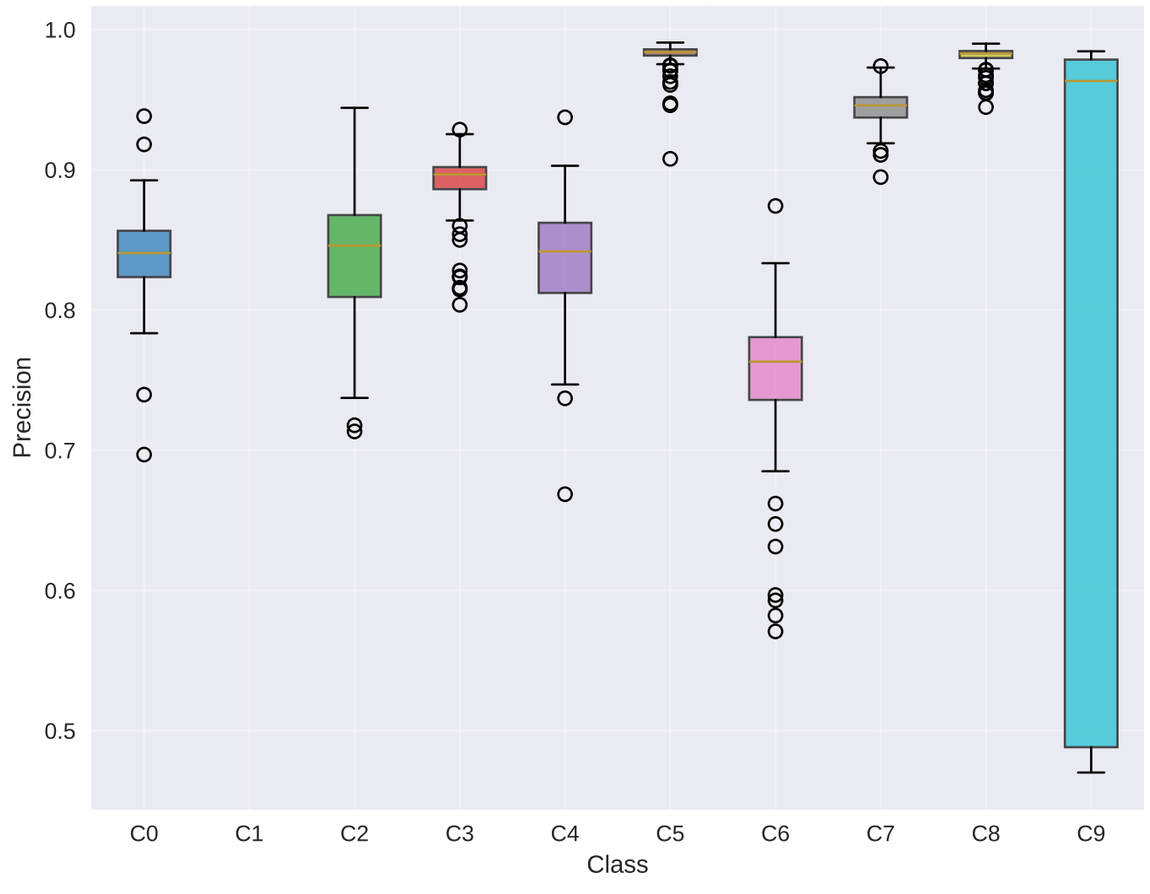


Per-Class Precision Analysis



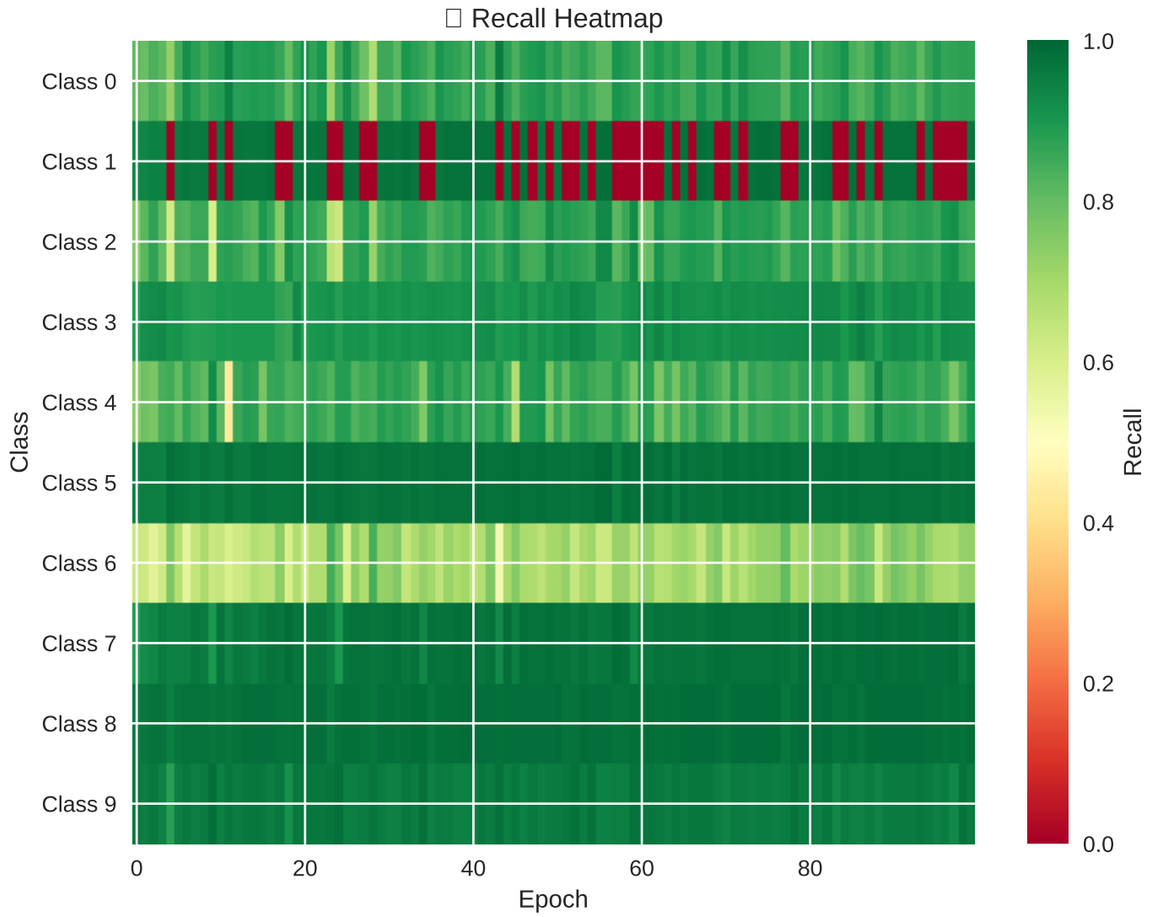


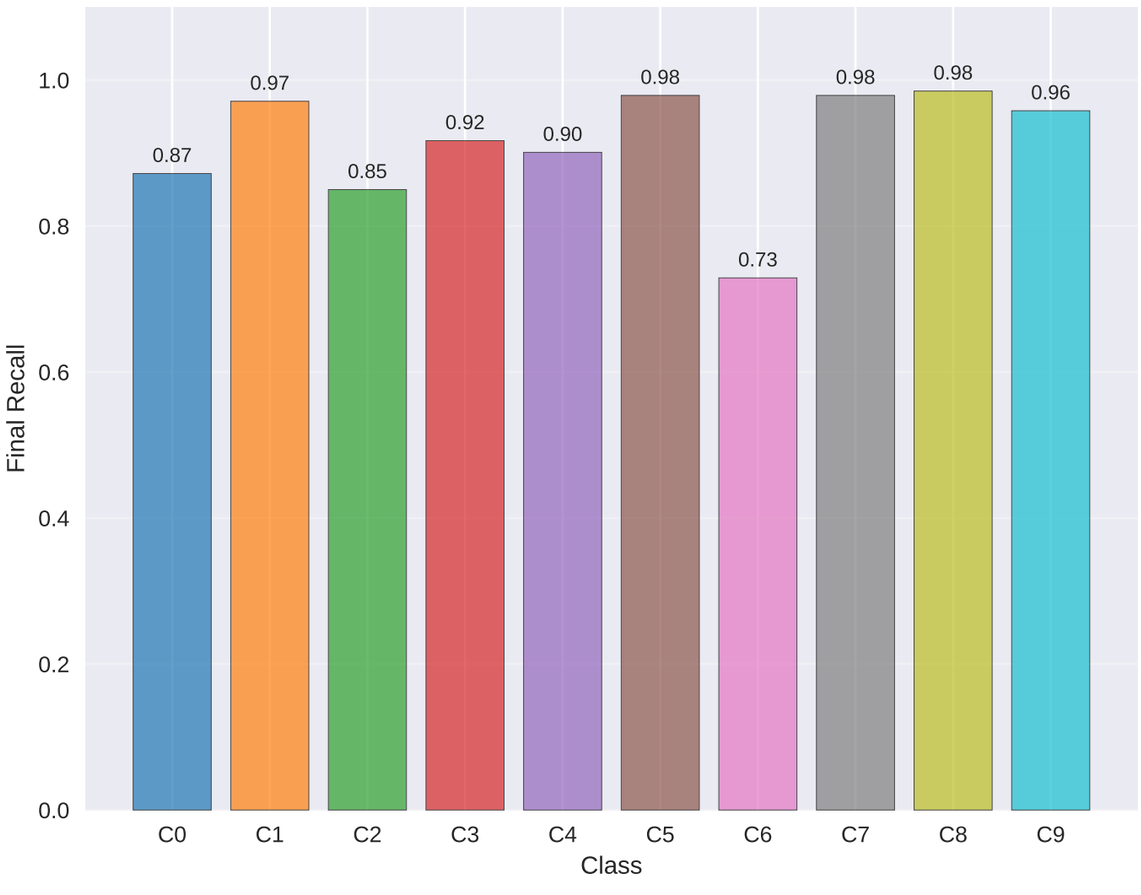


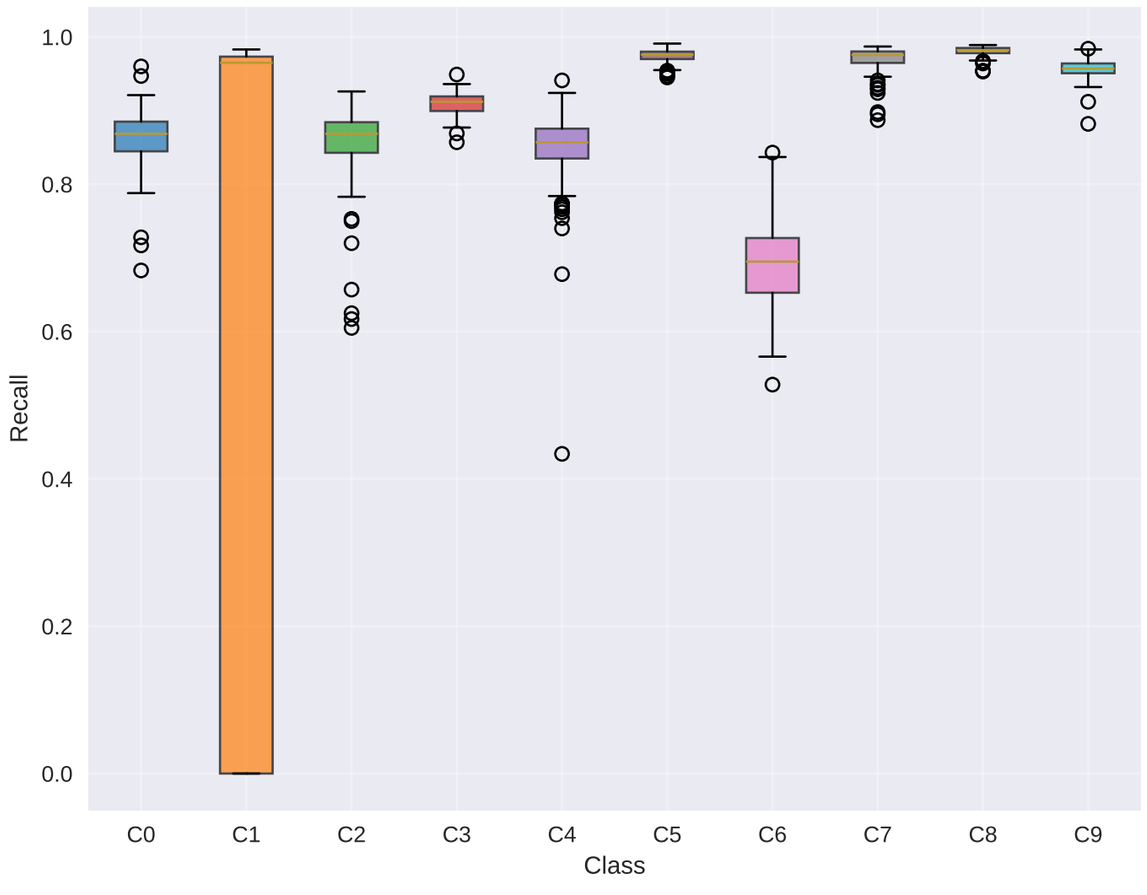


Per-Class Recall Analysis

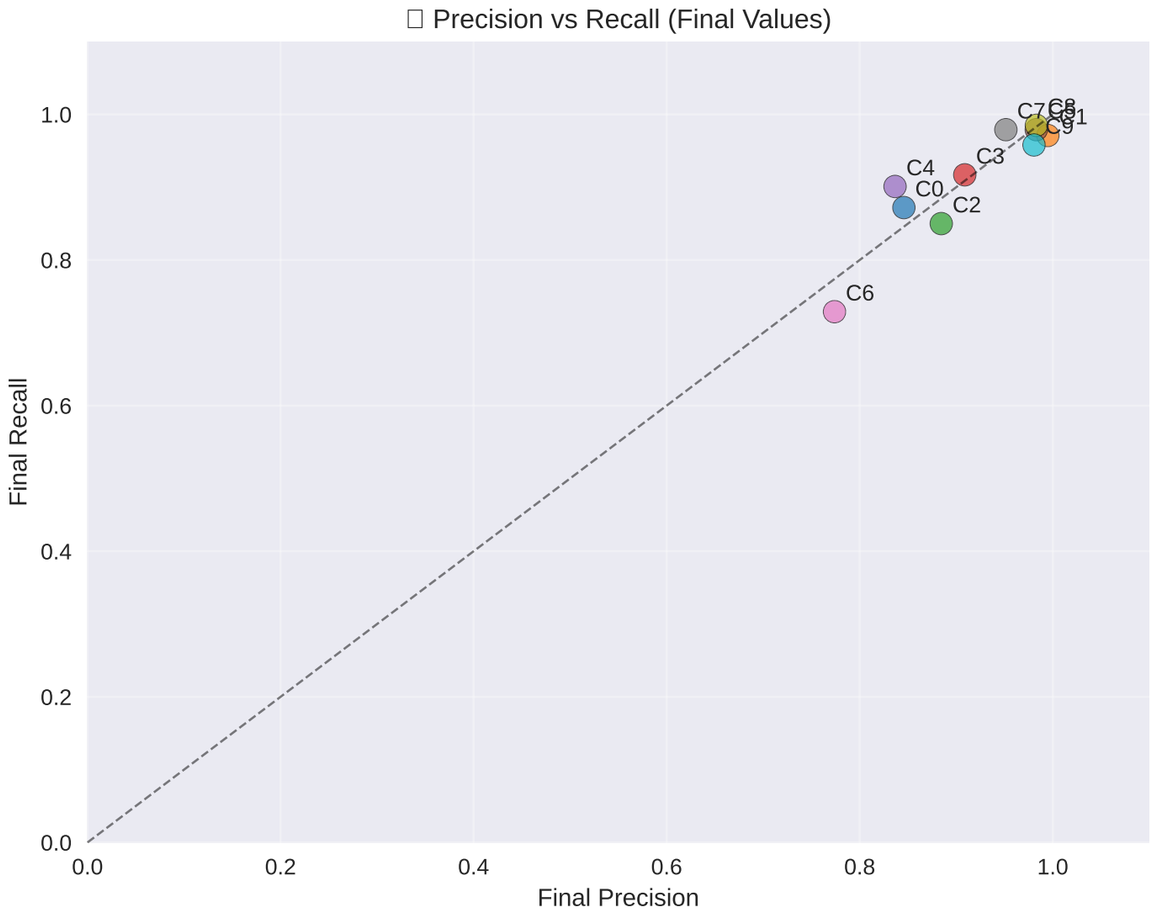


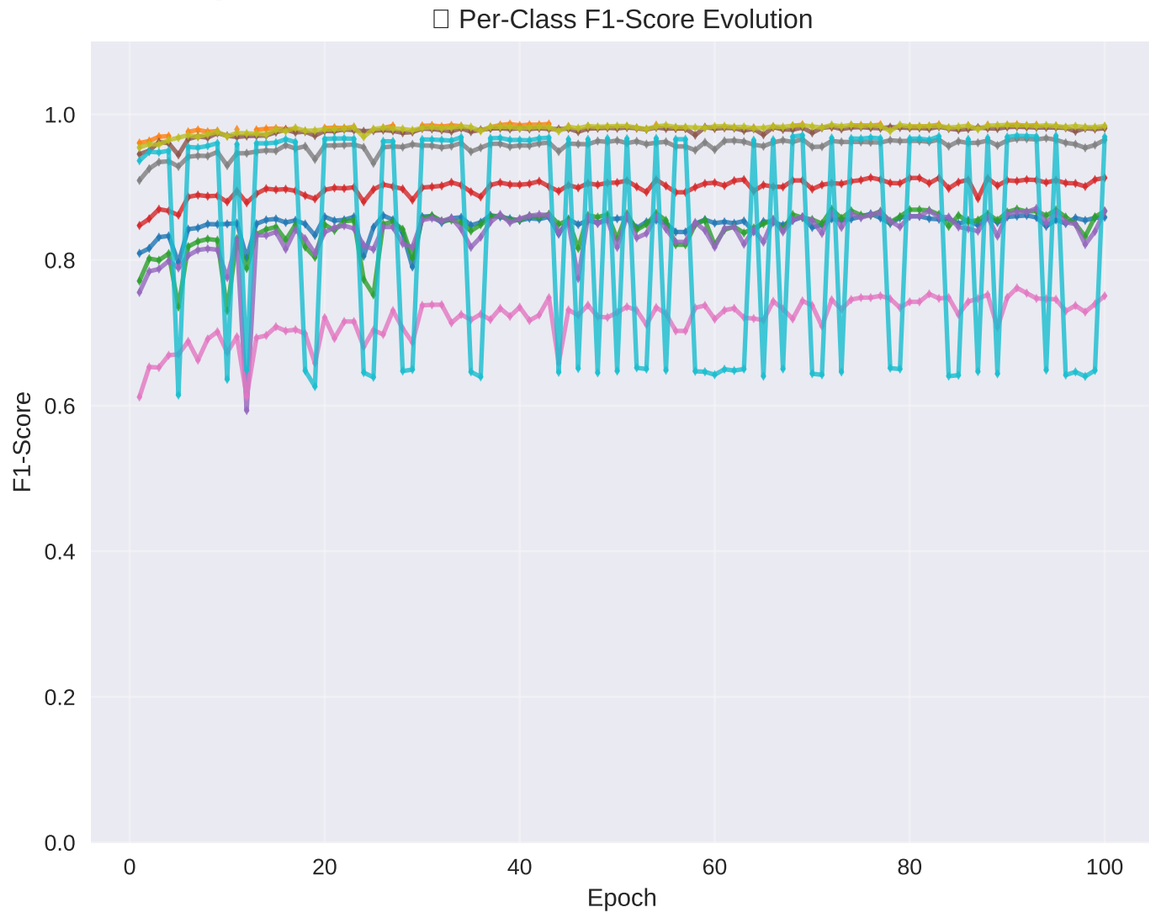


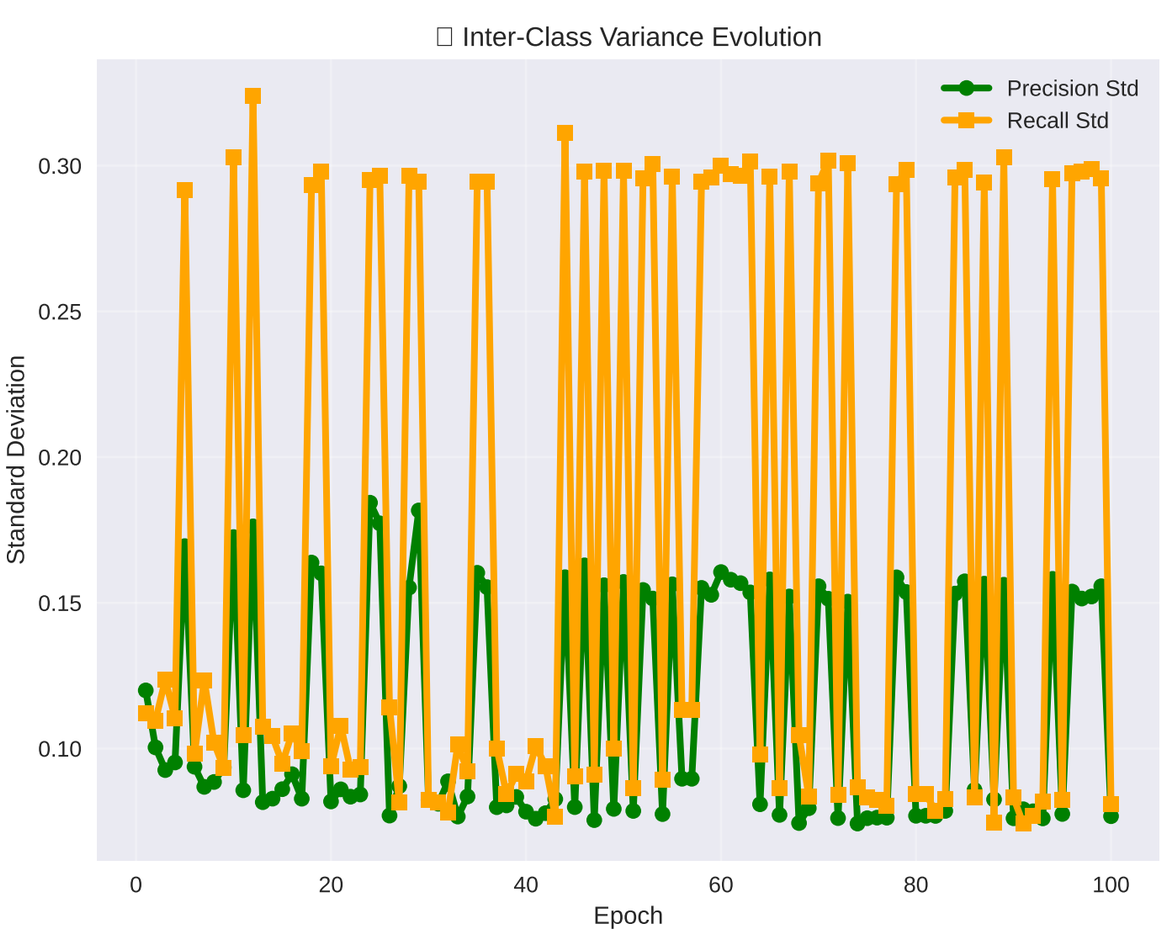


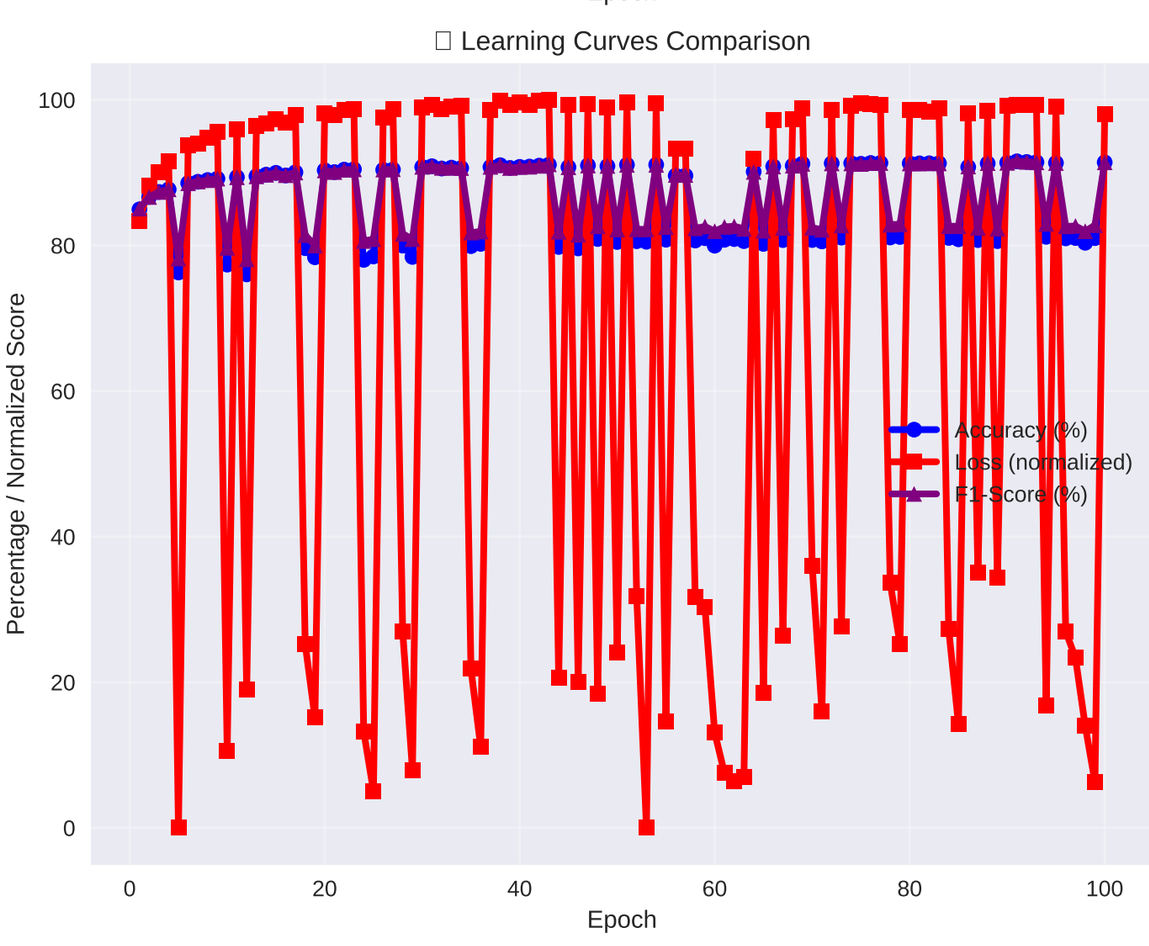


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-29 13:49:51
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.41%
• Loss: 2.7470
• Average Precision: 0.914
• Average Recall: 0.914
• Average F1-Score: 0.914

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.46%
• Riduzione Loss: +1.5182
• Stabilità Accuracy (CV): 0.0606

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.995)
• Peggiore Precision: Classe 6 (0.774)
• Migliore Recall: Classe 8 (0.985)
• Peggiore Recall: Classe 6 (0.729)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.846, R=0.872, F1=0.859
• Classe 1: P=0.995, R=0.971, F1=0.983
• Classe 2: P=0.884, R=0.850, F1=0.867
• Classe 3: P=0.909, R=0.917, F1=0.913
• Classe 4: P=0.837, R=0.901, F1=0.868
• Classe 5: P=0.983, R=0.979, F1=0.981
• Classe 6: P=0.774, R=0.729, F1=0.751
• Classe 7: P=0.951, R=0.979, F1=0.965
• Classe 8: P=0.983, R=0.985, F1=0.984
• Classe 9: P=0.981, R=0.958, F1=0.969

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: INSTABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Migliorare stabilità con learning rate scheduling
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